

DARSH DESAI

📍 Portsmouth, UK & St. Léger, CH

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PROFILE

Mechanical engineer working at the intersection of robotics, automation and intelligent manufacturing, where mechanical design meets control, sensing and autonomy: building things that move, sense and think.

Currently leading a 30-person team building a programmable Mars rover, with funding secured to start a LiDAR navigation robot. Wrote an RRT path planner for a nonholonomic robot, modelled ECU-based ABS control in Simulink, and turned solid-mechanics theory into a validated MATLAB tool that scored 100%. Committed to designs that survive contact with reality, built, tested, refined, and documented when they break.

CORE SKILLS

Robotics & Automation	Path planning (RRT / Dubins), autonomous mobile robots, control systems, Industry 4.0	Mechanical Design	PTC Creo, Fusion 360, AutoCAD, Onshape, Geometric Dimensioning & Tolerancing, DFM
Simulation & Analysis	MATLAB, Simulink, FEA, mould-flow analysis, SFD / BMD	Manufacturing	3D printing (FDM), injection moulding & tooling, CAM / CNC, laser cutting
Programming & Data	Python, VBA, MS Excel	Leadership	30-person team leadership, project & timeline management, onboarding, stakeholder alignment

SELECTED PROJECTS

Bridge Stress Analysis App

100%

MATLAB · App Designer

Automated shear-force and bending-moment tool for a simply supported bridge under Tandem and Lorry loads, with a validation layer that flags out-of-range outputs before they reach the user.

Modular Expandable Bag

82%

Fusion 360 · Team of 4

Led Pugh-matrix concept selection and designed a spring-loaded, tool-free modular attachment mechanism; owned its manufacture and validated it for load-bearing and usability.

ABS Quarter-Car Model

78%

MATLAB Simulink

Evaluated ECU / PLC control architectures and modelled bang-bang relay control with a hysteresis band to hold wheel slip at its optimal, peak-grip threshold.

RRT Path Planning

88%

MATLAB · Dubins state space

Dubins-based RRT planner for a nonholonomic robot, extended with greedy shortcutting and smoothing to produce shorter, collision-free paths across custom occupancy maps.

Multi-Stage Gearbox

81%

First-principles design · Team of 2

3:1 speed-increase gearbox for a 35 kW shipborne drive; belt, gear, shaft and bearing calculations to a 22,500-hour life, costed at £18,805 for a 100-unit run.

Injection-Moulded Bowl

71%

PTC Creo · CAM

Food-safe children's bowl taken to a two-part mould and CNC-simulated machining plan; compared PP and PETG to eliminate weld lines and air traps.

EXPERIENCE

Manufacturing Club Lead

2025 - Present

University of Portsmouth, UK

- ◆ Lead a 30-member multidisciplinary team designing and manufacturing a programmable Mars rover replica, directing mechanical design, fabrication and electrical integration.
- ◆ Manage a 6-member sub-team developing an autonomous mobile robot, overseeing LiDAR and Raspberry Pi integration for object detection and navigation.
- ◆ Secured university funding and institutional clearance for the robot; procure components with suppliers and lecturers within budget and to specification.
- ◆ Deliver Creo and 3D-printing workshops and advise the team on load-bearing components, manufacturability and system integration.

IT Industrialisation Intern

Aug - Sep 2025

Philip Morris International, Lausanne, Switzerland

- ◆ Worked in the department responsible for launching factories and digitally integrating production systems.
- ◆ Restructured the team's Confluence repository and documented the staged G0-G7 project lifecycle, improving organisation, searchability and onboarding.

Data Quality & Integration Intern

Jul - Aug 2025

Haut-Lac International Bilingual School, Switzerland

- ◆ Reconciled and validated data across HubSpot (CRM) and Engage against source records, running audits and system tests to resolve inconsistencies and redundancies.

IT Tech & Data Migration Intern

May - Aug 2024

Haut-Lac International Bilingual School, Switzerland

- ◆ Migrated 500+ student records between HubSpot and Engage, standardising formats and building report templates and secure extraction processes.
- ◆ Trained staff on both systems and developed and debugged VBA macros to fix formatting issues and improve data integrity.

Sheet Metal Worker Intern

May - Jun 2022

Car-Point Carrosseries SA, Switzerland

- ◆ Assisted a senior technician with vehicle body repair (dent correction, panel replacement and post-repair alignment) to industry-standard methods, with zero additional damage.

EDUCATION

BEng (Hons) Mechanical Engineering

2024 - 2028

University of Portsmouth, UK · On track for a First · IMechE-accredited

Engineering Mathematics & Numerical Analysis **97%**

Robotics & Automation **84%**

Engineering Design **82%**

Solid Mechanics & Dynamics **74%**

Thermodynamics & Fluid Mechanics **70%**

Computer-Aided Engineering & Manufacturing **68%**

IB Diploma

2017 - 2024

Haut-Lac International Bilingual School, Switzerland

Higher Level: Mathematics Analysis & Approaches, Physics, Business Management.

Standard Level: Computer Science, English Language & Literature, French B.

CERTIFICATIONS & INTERESTS

CERTIFICATIONS

First Aid (2023) · DELF B1 French (2021) · DELE A2 Spanish (2020)

INTERESTS

Cricket, University of Portsmouth CC (BUCS) · Personal robotics builds integrating mechanical design, electronics and control